

NAME : Shri Harshini Devi . A
REG.NO : 192524104
COURSE CODE : DSA0602
COURSE NAME : Data handling and Visualization

Class Activity: 10.07.2026

Scenario 1: Student Academic Performance Dashboard

CODE:

```
# Student Academic Performance Dashboard
```

```
# Create dataset
```

```
student <- data.frame(
```

```
  StudentID = c(101,102,103,104,105,106,107,108),
```

```
  Attendance = c(92,85,78,95,88,81,90,76),
```

```
  Marks = c(88,75,68,94,82,70,90,65),
```

```
  Department = c("CSE","AI","IT","ECE","CSE","AI","IT","ECE")
```

```
)
```

```
# Load ggplot2 package
```

```
library(ggplot2)
```

```
# Scatter Plot
```

```
ggplot(student, aes(x = Attendance, y = Marks, color = Department)) +  
  geom_point(size = 3) +  
  labs(  
    title = "Student Academic Performance Dashboard",  
    x = "Attendance (%)",  
    y = "Internal Marks",  
    color = "Department"  
  ) +  
  theme_minimal()
```

OUTPUT:



Scenario 2: Electric Vehicle Market Analysis

CODE:

```
# Load ggplot2 package
```

```
library(ggplot2)
```

```
# Create dataset
```

```
ev <- data.frame(
```

```
  VehicleModel = c("Model A", "Model B", "Model C", "Model D", "Model E"),
```

```
  BatteryCapacity = c(40, 50, 60, 75, 90),
```

```
  Price = c(12, 16, 20, 28, 35),    # Price in Lakhs
```

```
  DrivingRange = c(250, 320, 400, 480, 550), # Range in km
```

```
  Manufacturer = c("Tata", "MG", "Mahindra", "Hyundai", "BYD")
```

```
)
```

```
# Bubble Chart
```

```
ggplot(ev, aes(x = BatteryCapacity,
```

```
  y = Price,
```

```
  size = DrivingRange,
```

```
  color = Manufacturer)) +
```

```
geom_point(alpha = 0.8) +
```

```
labs(
```

```
  title = "Electric Vehicle Market Analysis",
```

```
  x = "Battery Capacity (kWh)",
```

```
  y = "Price (Lakhs)",
```

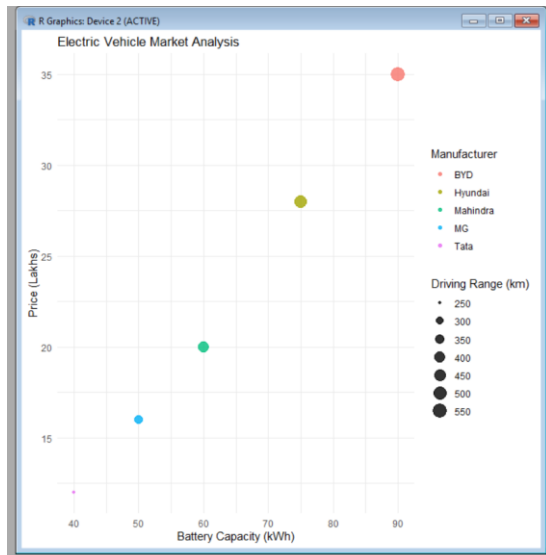
```
  size = "Driving Range (km)",
```

```
  color = "Manufacturer"
```

```
) +
```

```
theme_minimal()
```

OUTPUT:



Scenario 3: Hospital Patient Admission Analysis

CODE:

```
# Load ggplot2 package
```

```
library(ggplot2)
```

```
# Create dataset
```

```
hospital <- data.frame(
```

```
  Department = c("Cardiology", "Neurology", "Orthopedics", "Pediatrics",  
  "General"),
```

```
  Patients = c(120, 95, 80, 140, 160),
```

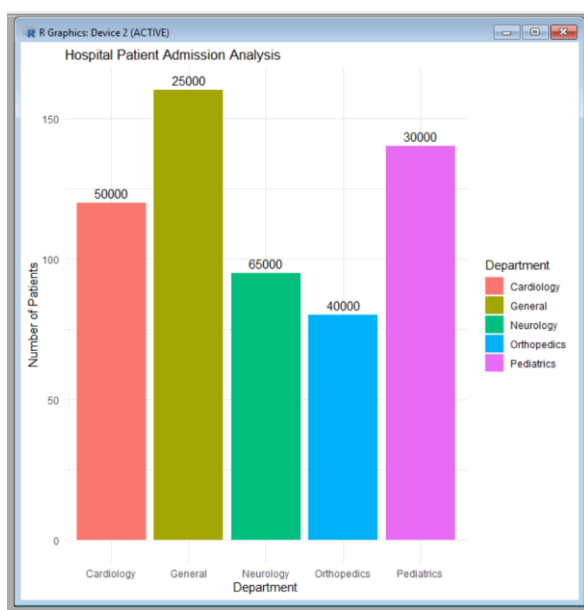
```
  TreatmentCost = c(50000, 65000, 40000, 30000, 25000)
```

)

Bar Chart

```
ggplot(hospital, aes(x = Department, y = Patients, fill = Department)) +  
  geom_bar(stat = "identity") +  
  geom_text(aes(label = TreatmentCost), vjust = -0.5, size = 4) +  
  labs(  
    title = "Hospital Patient Admission Analysis",  
    x = "Department",  
    y = "Number of Patients",  
    fill = "Department"  
  ) +  
  theme_minimal()
```

OUTPUT:



Scenario 4: Weather Monitoring System

CODE:

```
# Load ggplot2 package
```

```
library(ggplot2)
```

```
# Create dataset
```

```
weather <- data.frame(
```

```
  Date = rep(1:5, 3),
```

```
  Temperature = c(30, 31, 32, 33, 34,
```

```
                 28, 29, 30, 31, 32,
```

```
                 26, 27, 28, 29, 30),
```

```
  City = rep(c("Chennai", "Bengaluru", "Delhi"), each = 5)
```

```
)
```

```
# Line Chart
```

```
ggplot(weather,
```

```
  aes(x = Date,
```

```
      y = Temperature,
```

```
      color = City,
```

```
      linetype = City,
```

```
      group = City)) +
```

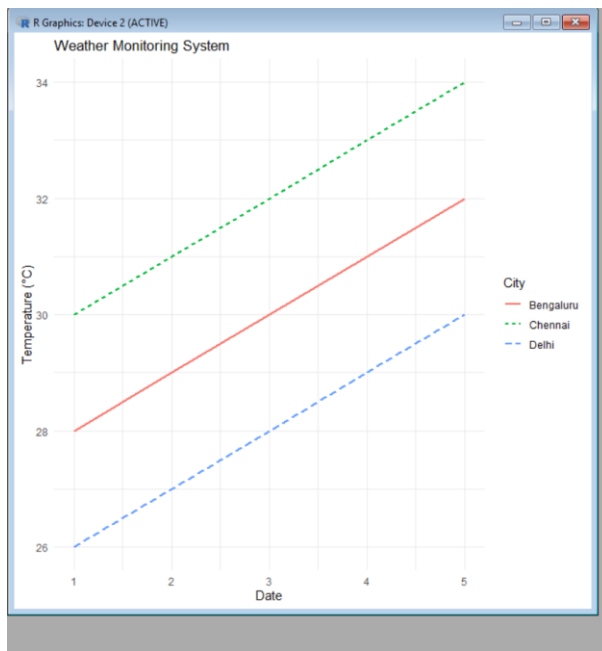
```
geom_line(size = 1) +
```

```
labs(
```

```
  title = "Weather Monitoring System",
```

```
x = "Date",  
y = "Temperature (°C)",  
color = "City",  
linetype = "City"  
) +  
theme_minimal()
```

OUTPUT:



Scenario 5 : Supermarket Sales Analysis

CODE:

```
# Load ggplot2 package  
library(ggplot2)  
  
# Create dataset  
sales <- data.frame(
```

```

Month = c("Jan","Jan","Jan","Jan","Feb","Feb","Feb","Feb"),
ProductCategory = c("Groceries","Groceries","Beverages","Beverages",
                     "Groceries","Groceries","Beverages","Beverages"),
SalesAmount = c(50000,45000,30000,35000,55000,48000,32000,38000),
Branch = c("Branch A","Branch B","Branch A","Branch B",
           "Branch A","Branch B","Branch A","Branch B")
)

```

Grouped Bar Chart

```

ggplot(sales,
       aes(x = ProductCategory,
           y = SalesAmount,
           fill = Branch)) +
  geom_bar(stat = "identity", position = "dodge") +
  facet_wrap(~Month) +
  labs(
    title = "Supermarket Sales Analysis",
    x = "Product Category",
    y = "Sales Amount",
    fill = "Branch"
  ) +
  theme_minimal()

```

OUTPUT:



Scenario 6: Employee Salary Distribution

CODE:

```
# Load ggplot2 package
```

```
library(ggplot2)
```

```
# Create dataset
```

```
employee <- data.frame(
```

```
  EmployeeID = 1:10,
```

```
  Salary = c(25000, 30000, 35000, 40000, 45000,
```

```
            50000, 55000, 60000, 65000, 70000),
```

```
  Experience = c(1,2,3,4,5,6,7,8,9,10),
```

```
  Department = c("HR","IT","Finance","Sales","HR",
```

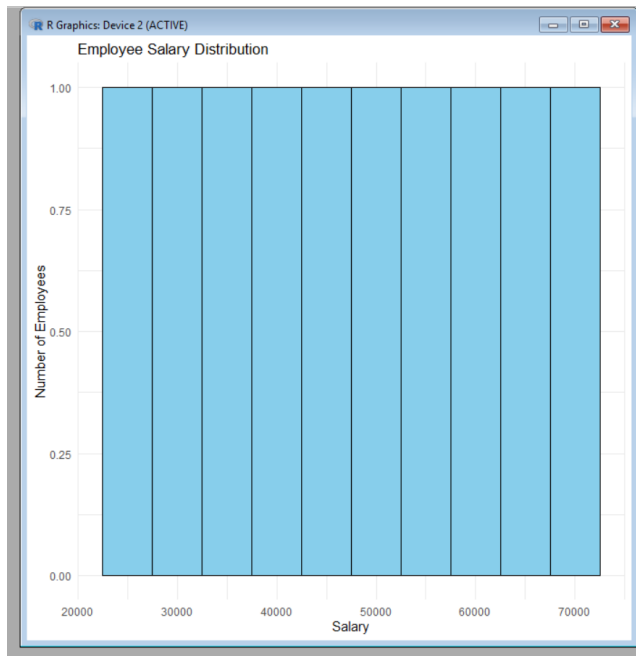
```
    "IT","Finance","Sales","HR","IT")
)
```

```
# Histogram (Bin width = 5000)
ggplot(employee, aes(x = Salary)) +
  geom_histogram(binwidth = 5000,
    fill = "skyblue",
    color = "black") +
  labs(
    title = "Employee Salary Distribution",
    x = "Salary",
    y = "Number of Employees"
  ) +
  theme_minimal()
```

```
# Histogram (Bin width = 10000)
ggplot(employee, aes(x = Salary)) +
  geom_histogram(binwidth = 10000,
    fill = "lightgreen",
    color = "black") +
  labs(
    title = "Employee Salary Distribution (Larger Bin Size)",
    x = "Salary",
    y = "Number of Employees"
  ) +
```

```
theme_minimal()
```

OUTPUT:



Scenario 7: Smartphone Market Comparison

CODE:

```
# Load ggplot2 package
```

```
library(ggplot2)
```

```
# Create dataset
```

```
phone <- data.frame(
```

```
  Brand = c("Samsung", "Apple", "OnePlus", "Xiaomi", "Realme"),
```

```
  RAM = c(8, 6, 12, 8, 4),
```

```
  ProcessorSpeed = c(2.8, 3.2, 3.0, 2.6, 2.4), # GHz
```

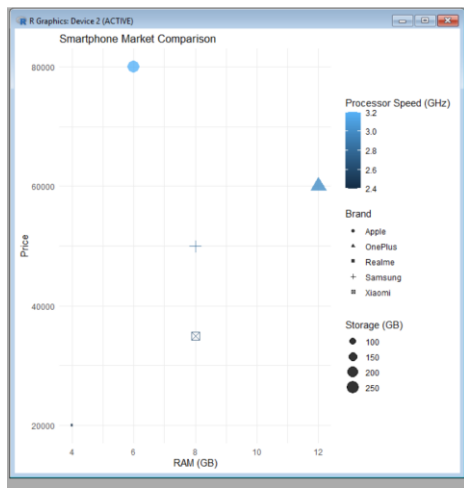
```
  Price = c(50000, 80000, 60000, 35000, 20000),
```

```
  Storage = c(128, 256, 256, 128, 64)
```

```
)
```

```
# Scatter Plot
ggplot(phone,
  aes(x = RAM,
      y = Price,
      shape = Brand,
      size = Storage,
      color = ProcessorSpeed)) +
  geom_point(alpha = 0.8) +
  labs(
    title = "Smartphone Market Comparison",
    x = "RAM (GB)",
    y = "Price",
    shape = "Brand",
    size = "Storage (GB)",
    color = "Processor Speed (GHz)"
  ) +
  theme_minimal()
```

OUTPUT:



Scenario 8: Online Movie Rating Analysis

CODE:

```
# Load ggplot2 package
```

```
library(ggplot2)
```

```
# Create dataset
```

```
movie <- data.frame(
```

```
  Genre = c("Action", "Comedy", "Drama", "Horror", "Romance"),
```

```
  AverageRating = c(4.5, 3.8, 4.7, 3.5, 4.2),
```

```
  NumberOfReviews = c(1200, 900, 1500, 700, 1000),
```

```
  ReleaseYear = c(2022, 2022, 2023, 2023, 2024)
```

```
)
```

```
# Heatmap
```

```
ggplot(movie,
```

```
  aes(x = factor(ReleaseYear),
```

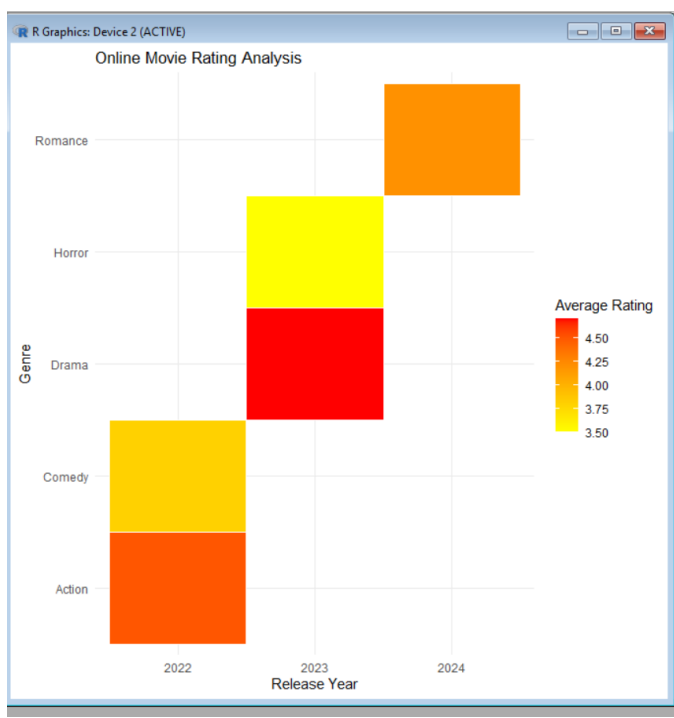
```
    y = Genre,
```

```

    fill = AverageRating)) +
  geom_tile(color = "white") +
  scale_fill_gradient(low = "yellow", high = "red") +
  labs(
    title = "Online Movie Rating Analysis",
    x = "Release Year",
    y = "Genre",
    fill = "Average Rating"
  ) +
  theme_minimal()

```

OUTPUT:



Scenario 9: COVID-19 State-wise Analysis

CODE:

```
# Load ggplot2 package
```

```
library(ggplot2)
```

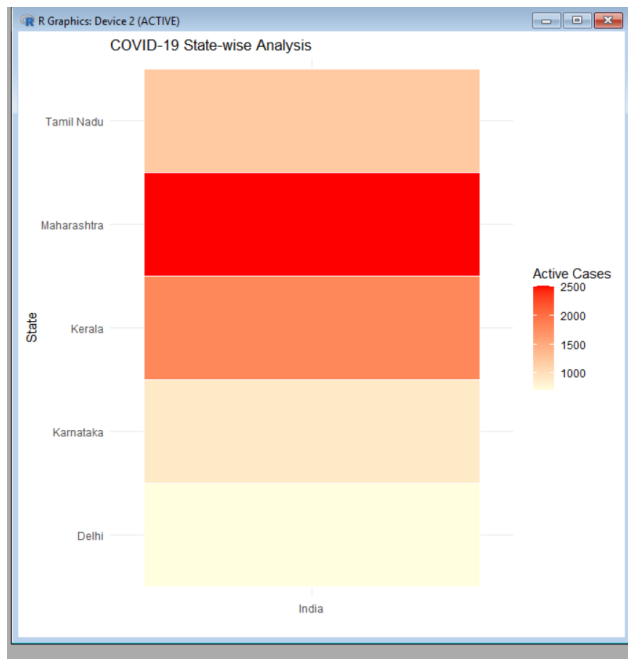
```
# Create dataset
```

```
covid <- data.frame(  
  State = c("Tamil Nadu", "Kerala", "Karnataka", "Maharashtra", "Delhi"),  
  ActiveCases = c(1200, 1800, 900, 2500, 700),  
  VaccinationPercentage = c(92, 90, 91, 89, 94),  
  Population = c(78000000, 35000000, 68000000, 124000000, 20000000)  
)
```

```
# Tile Plot (Heatmap)
```

```
ggplot(covid,  
  aes(x = "India",  
    y = State,  
    fill = ActiveCases)) +  
  geom_tile(color = "white") +  
  scale_fill_gradient(low = "lightyellow", high = "red") +  
  labs(  
    title = "COVID-19 State-wise Analysis",  
    x = "",  
    y = "State",  
    fill = "Active Cases"  
  ) +  
  theme_minimal()
```

OUTPUT:



Scenario 10: Iris Flower Classification

CODE:

```
# Load ggplot2 package
```

```
library(ggplot2)
```

```
# Use built-in Iris dataset
```

```
data(iris)
```

```
# Scatter Plot
```

```
ggplot(iris,  
  aes(x = Sepal.Length,  
    y = Petal.Length,  
    color = Species,  
    shape = Species,  
    size = Sepal.Width)) +
```



```
geom_point(alpha = 0.8) +  
labs(  
  title = "Iris Flower Classification",  
  x = "Sepal Length",  
  y = "Petal Length",  
  color = "Species",  
  shape = "Species",  
  size = "Sepal Width"  
) +  
theme_minimal()
```

OUTPUT:

